

1) Elaborate on the following terms

- i. **Flow control**
- ii. **RNR**
- iii. **Half duplex**
- iv. **SMTP**

Flow control

- ⇧ Ensure sending entity does not overwhelm receiving entity by preventing buffer overflow.
- ⇧ influenced by:
 - transmission time
 - time taken to emit all bits into medium
 - propagation time
 - time for a bit to traverse the link
- ⇧ Two methods:
 - o STOP AND WAIT:
In this source wait for the acknowledge before the next frame.
 - o SLIDING WINDOW:
Allows transmission of multiple frames if A is sender and B is receiver and both will be wanted by a duplex link.

RNR

- ⇧ Sliding window flow control is a technique used in data communication to manage the flow of data between the sender and receiver.
- ⇧ The receiver in sliding window flow control maintains a receive window, which represents the range of acceptable sequence numbers for incoming data.
- ⇧ When the receiver is not ready to accept additional data, it sends a Receiver-Not-Ready (RNR) frame to the sender.
- ⇧ The RNR frame indicates to the sender that the receiver's buffer is full or there is some other condition that prevents it from accepting more data.
- ⇧ Upon receiving an RNR frame, the sender must stop sending data until it receives a Receiver-Ready (RR) frame from the receiver.
- ⇧ The RNR condition allows the receiver to control the rate at which data is sent by signaling to the sender when it needs to pause transmission.

- ⇧ The sender maintains a send window, which represents the range of sequence numbers for which it has sent data but has not received acknowledgment.
- ⇧ When the sender receives an RNR frame, it stops sending data and waits for a RR frame from the receiver to resume transmission.
- ⇧ Once the receiver is ready to accept data again, it sends a RR frame to the sender, indicating that it can continue sending.
- ⇧ The RNR condition helps prevent data overflow at the receiver's end and ensures that the receiver can process the incoming data at its own pace.

Eg: RNR5 means receiver has received till 4th frame but not ready to receive next frame.

Half duplex

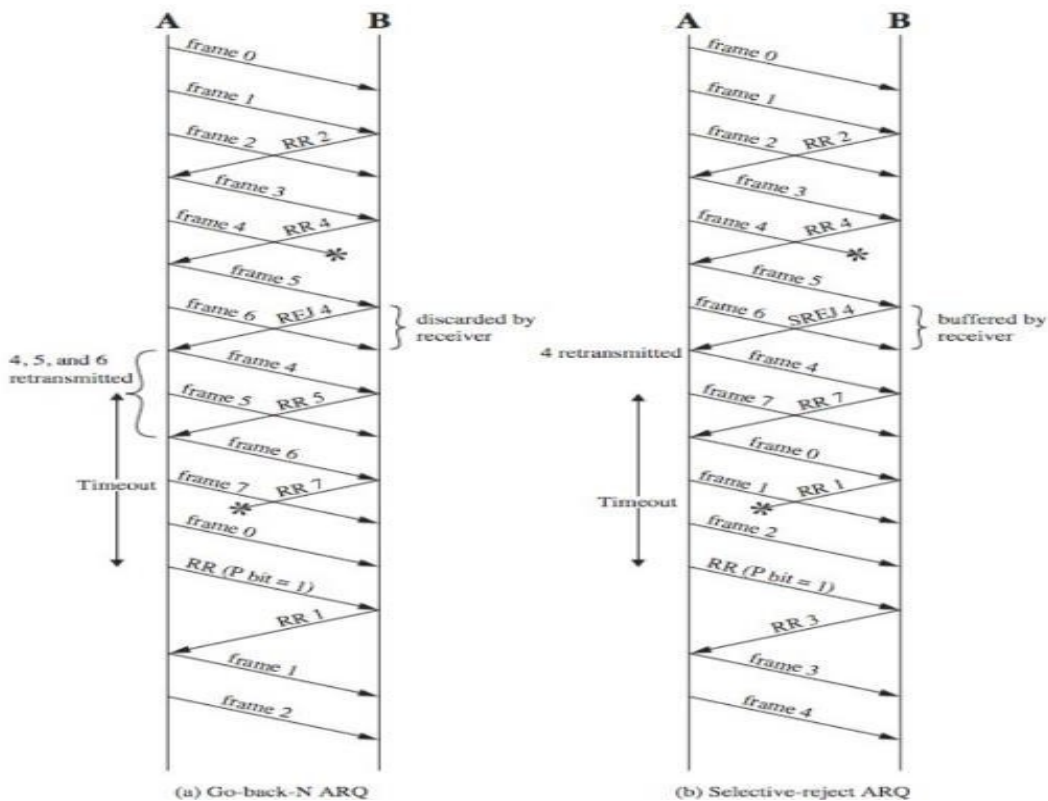
- ⇧ In computer networking, half-duplex refers to a communication mode where data can be transmitted in both directions, but not simultaneously.
- ⇧ In a half-duplex system, only one device can transmit data at a time, while the other device acts as a receiver. The roles can be switched between the devices.
- ⇧ Half-duplex communication can be contrasted with full-duplex communication, where data can be transmitted simultaneously in both directions.
- ⇧ In a half-duplex system, devices must take turns to transmit and receive data, resulting in a shared communication channel.
- ⇧ Half-duplex communication is commonly found in Ethernet networks using half-duplex hubs or shared media, such as coaxial cables.
- ⇧ In half-duplex Ethernet, the CSMA/CD (Carrier Sense Multiple Access with Collision Detection) algorithm is used to avoid collisions when multiple devices attempt to transmit simultaneously.
- ⇧ When a device wants to transmit in a half-duplex network, it first listens to the channel to check if it is currently being used by another device. If the channel is clear, it begins transmitting.
- ⇧ In half-duplex mode, collisions can occur if two or more devices try to transmit data at the same time. When a collision is detected, the devices involved must stop transmitting and wait for a random time before attempting to retransmit.
- ⇧ Due to the limitations of half-duplex communication, it is generally less efficient than full-duplex communication since it introduces delays and reduces the overall throughput of the network.

- ⇧ Half-duplex communication is commonly used in scenarios where the cost or complexity of implementing full-duplex communication is prohibitive, such as walkie-talkies, certain wireless networks, and some legacy Ethernet networks.

SMTP

- ⇧ SMTP is a widely used protocol for sending email messages over the Internet.
- ⇧ SMTP operates on the application layer of the TCP/IP protocol stack and uses port number 25 by default.
- ⇧ The primary function of SMTP is to transfer email messages from the sender's mail server to the recipient's mail server.
- ⇧ SMTP follows a client-server architecture, where the sender's email client acts as the SMTP client, and the recipient's mail server acts as the SMTP server.
- ⇧ SMTP uses a store-and-forward mechanism, where the sending SMTP server temporarily stores the email message and forwards it to the next SMTP server until it reaches the recipient's mail server.
- ⇧ SMTP supports both text and binary data transmission, allowing email attachments to be included in messages.
- ⇧ The SMTP conversation between the client and server typically involves a series of commands and responses. Common commands include HELO/EHLO (to initiate the session), MAIL FROM (to specify the sender's email address), RCPT TO (to specify the recipient's email address), and DATA (to send the email message content).
- ⇧ SMTP also provides error handling mechanisms, including status codes and error messages, to indicate the success or failure of the email transmission.
- ⇧ SMTP is a reliable protocol that ensures the delivery of email messages by employing various mechanisms, such as retries, queuing, and error notifications.
- ⇧ While SMTP is primarily used for sending emails, additional protocols like POP (Post Office Protocol) or IMAP (Internet Message Access Protocol) are used to retrieve emails from a mail server by the email client

2) With diagram, distinguish between Go-Back-N-ARQ and selective reject ARQ.



Go Back N ARQ	Selective Reject ARQ
1) When the receiver finds a segment in error, it tells the transmitter to go back, resend that segment and all succeeding segments.	1) The receiver, upon finding a errored segment, will request that segment be resent and only that segment is resent.
2) May have to retransmit good segments.	2) Only retransmit lost or errored segments.

3) Easy to implement.	3) Requires more powerful processing and more buffer space.
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Explain in detail GO-Back-N-ARQ.

- *GO-back-N-ARQ is based on sliding window technique.
- *Go-Back-N ARQ, where **N** is the sender's window size. Suppose we say that Go-Back-3, which means that the three frames can be sent at a time before expecting the acknowledgment from the receiver.
- *If no error, then the Acknowledgement will be as usual.
- *It uses window to control number of outstanding frames.
- *If there are errors, then it will reply with rejection.
- *It will discard that frame and all future frames until error frame will be received correctly.
- *Transmitter must go back and retransmit that frame and all subsequent frames.

GO Back N Handling

***Damaged Frame**

- ¬ Error in frame i so receiver rejects frame i .
- ¬ Transmitter retransmits frame from i .

***Lost Frame**

- ¬ Frame i lost and either
 - Transmitter sends $i+1$ and receiver gets frame $i+1$ out of sequence and rejects frame i .
 - Or transmitter times out and sends Ack(acknowledgement) with P bit set which receiver responds to with ACK i .
- ¬ Transmitter then retransmits frames from i .

***Damaged Acknowledgement**

- ¬ Receiver gets frame i , sends ack($i+1$) which is lost.
- ¬ ACK's are cumulative, so next ACK ($i+n$) may arrive before transmitter times out on frame i .
- ¬ If the transmitter times out, it sends ACK with P bit set.
- ¬ It can be repeated a number of times before a reset procedure is initiated.

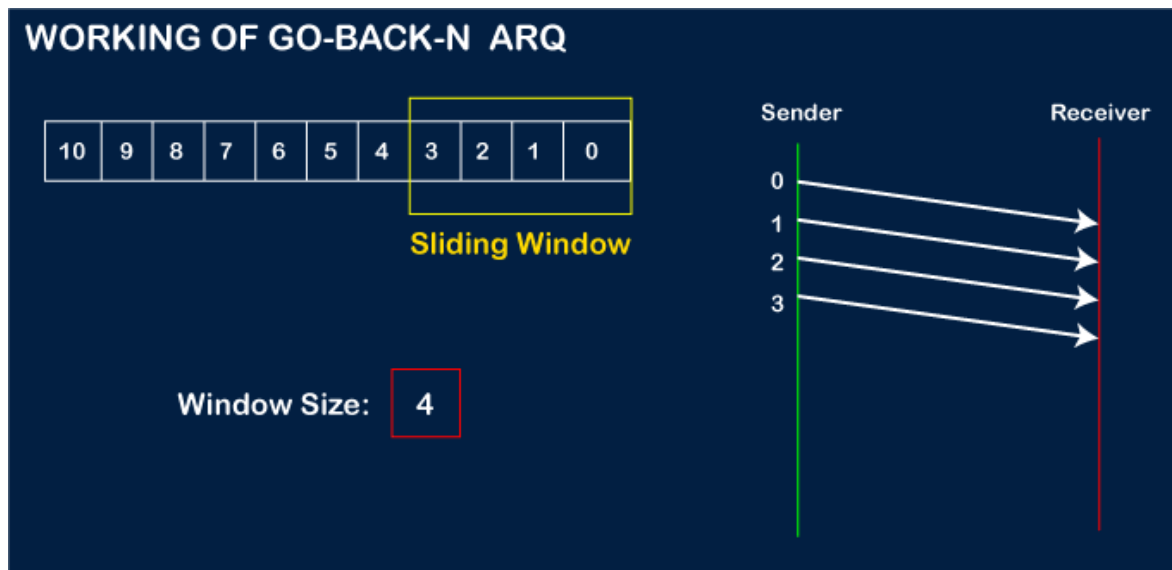
***Damaged Rejection**

- ¬ Rejects for the damaged frame is lost.
- ¬ It is handled for lost frame when transmitter times out.

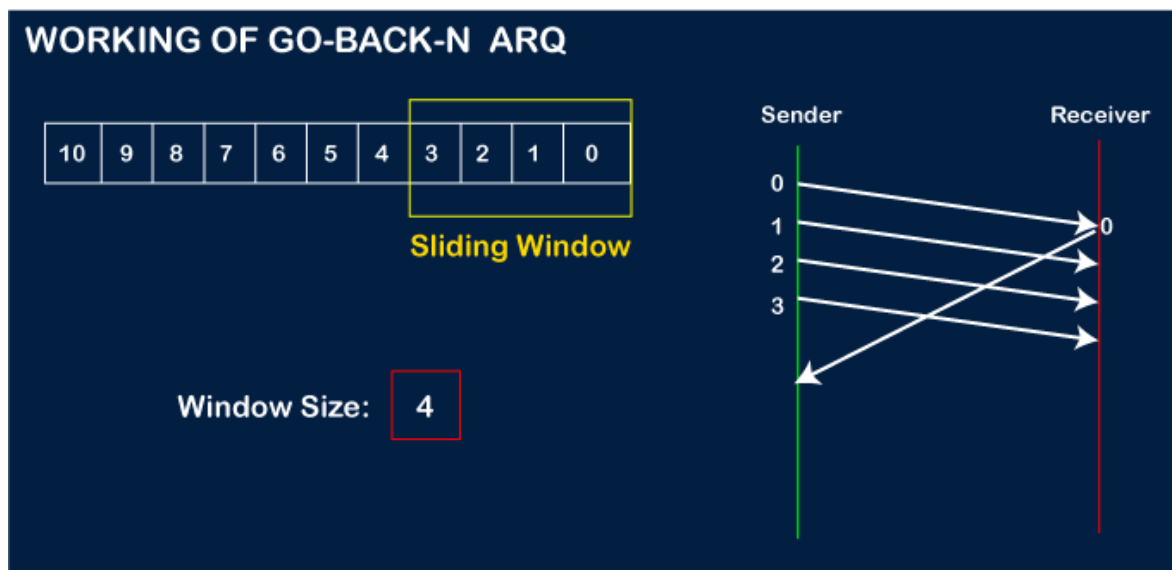
Working of Go-Back-N ARQ

Suppose there are a sender and a receiver, and let's assume that there are 11 frames to be sent. These frames are represented as 0,1,2,3,4,5,6,7,8,9,10, and these are the sequence numbers of the frames.

Step 1: Firstly, the sender will send the first four frames to the receiver, i.e., 0,1,2,3, and now the sender is expected to receive the acknowledgment of the 0th frame.

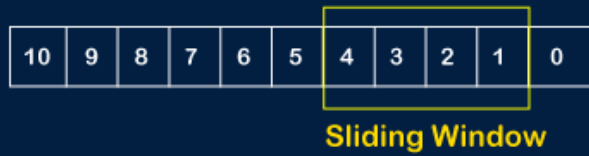


Let's assume that the receiver has sent the acknowledgment for the 0 frame, and the receiver has successfully received it.

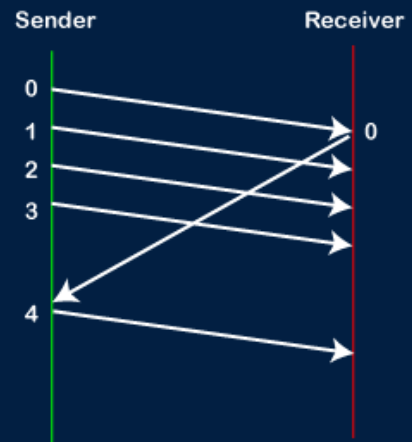


The sender will then send the next frame, i.e., 4, and the window slides containing four frames (1,2,3,4).

WORKING OF GO-BACK-N ARQ

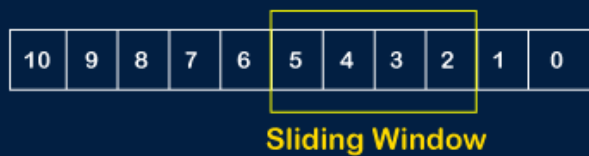


Window Size: 4

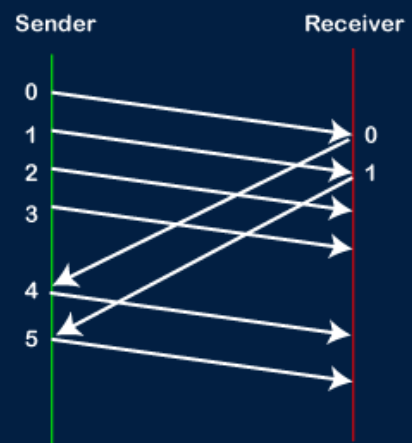


The receiver will then send the acknowledgment for the frame no 1. After receiving the acknowledgment, the sender will send the next frame, i.e., frame no 5, and the window will slide having four frames (2,3,4,5).

WORKING OF GO-BACK-N ARQ

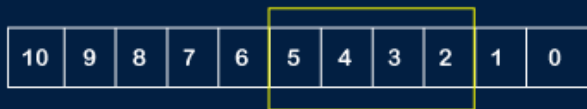


Window Size: 4



Now, let's assume that the receiver is not acknowledging the frame no 2, either the frame is lost, or the acknowledgment is lost. Instead of sending the frame no 6, the sender Go-Back to 2, which is the first frame of the current window, retransmits all the frames in the current window, i.e., 2,3,4,5.

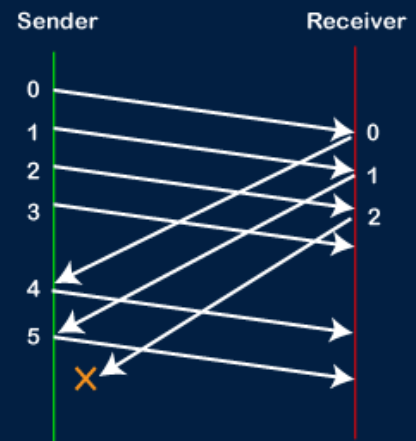
WORKING OF GO-BACK-N ARQ



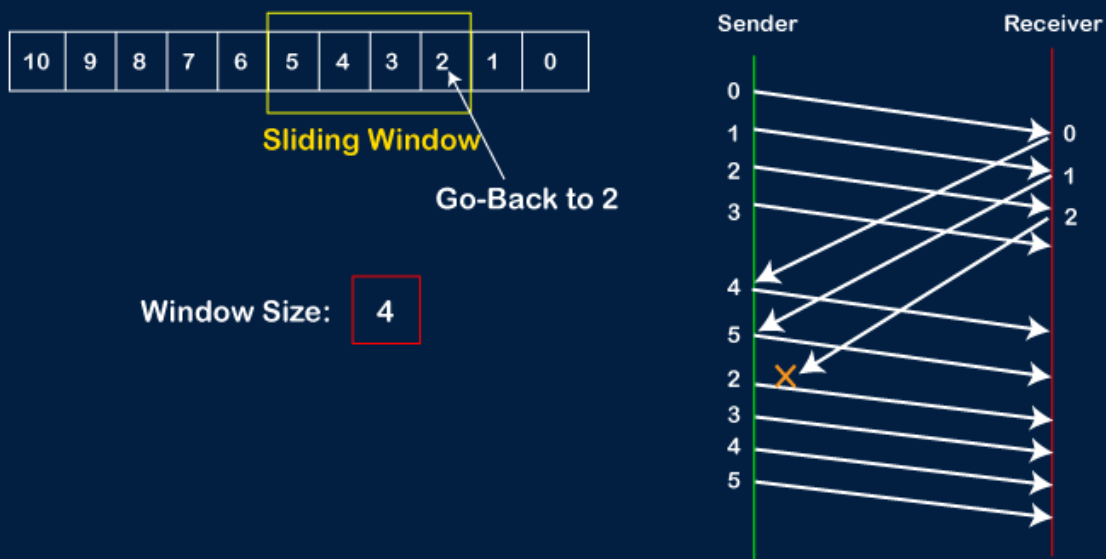
Sliding Window

Window Size:

4

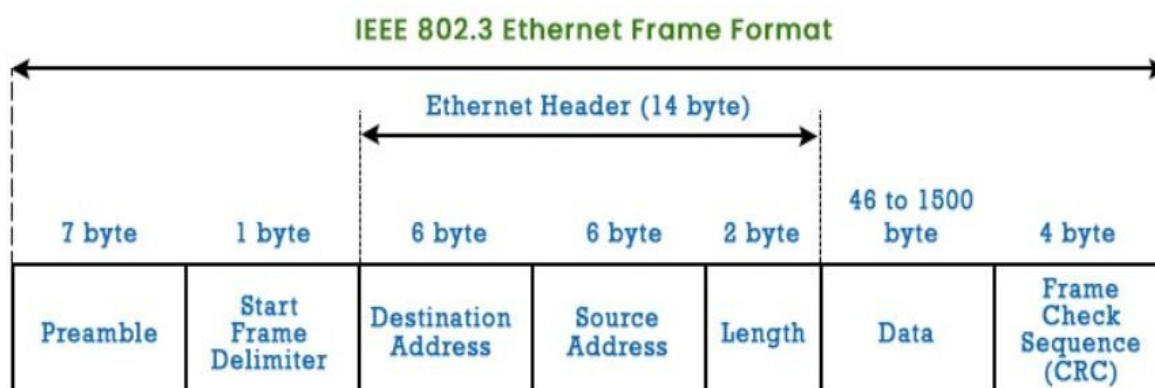


WORKING OF GO-BACK-N ARQ



4.) Describe the frame structure of IEEE 802.3 Ethernet.

The IEEE 802.3 standard defines the fundamental frame format that is necessary for all MAC implementations. However, the core functionality of the protocol is being extended by several optional forms. Preamble and SFD, which operate at the physical layer, begin an Ethernet frame. The packet's payload follows the Ethernet header, which includes the MAC addresses for the source and destination. CRC, the final field, is utilized to find errors. Let's now examine each section of the fundamental frame format.



1. **PREAMBLE** - Ethernet frames begin with a 7-byte. This is a sequence of alternate 0s and 1s that denotes the beginning of the frame and enables bit synchronization between the sender and receiver.

PRE (Preamble) was initially developed to accommodate the loss of a few bits as a result of signal delays. However, the frame bits in high-speed Ethernet today are protected without the need for a preamble.

Prior to the actual frame beginning, PRE (Preamble) alerts the receiver that a frame is about to start and enables the receiver to lock onto the data stream.

2. **Start of frame Delimiter (SFD)** - This 1-byte field is always set to **10101011**. The destination address is the next set of bits that will begin the frame, as indicated by SFD. The preamble is frequently referred to as 8 Bytes since SFD is sometimes seen as a component of PRE. The SFD notifies the station or stations that synchronization is now impossible.
3. **Destination Address** - This 6-Byte element contains the MAC address of the device for which the data is intended.
4. **Source Address** - This 6-byte element contains the source machine's MAC address. Since Source Address is always a unique address (Unicast), 0 is always the least significant bit of the first byte.
5. **Length** - A 2-Byte field called Length represents the size of an Ethernet frame as a whole. Due to some inherent constraints of Ethernet, this 16-bit field can store length values from 0 to 65534, but length values greater than 1500 are not permitted.
6. **Data** - This area, sometimes referred to as the Payload, is where the real data is placed. If Internet Protocol is utilised via Ethernet, both the IP header and data will be placed here. The longest possible piece of data might be 1500 bytes long. If the data length is less than the minimum length, which is 46 bytes, padding 0's are appended to make up the difference.
7. **Cyclic Redundancy Check (CRC)** - CRC is a field of 4 bytes. The data in this field is a 32-bit hash code created using the fields for the destination address, source address, length, and data. Data is damaged if the checksum calculated by the destination differs from the checksum value supplied.

5. What is flow control? Explain sliding window flow control with neat diagram.

Flow control is a technique used to regulate data transfer between computers or other nodes in a network. Flow control ensures that the sending entity does not overwhelm the receiving entity.

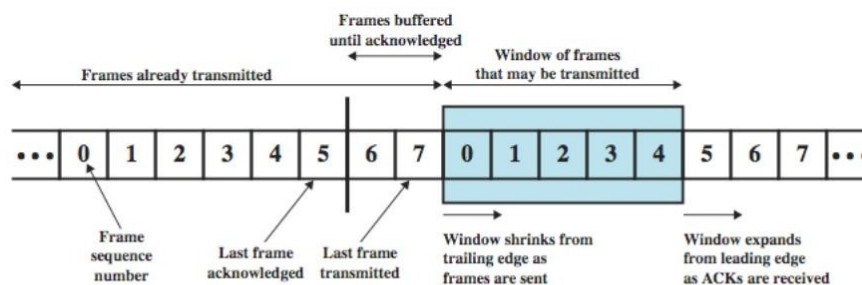
Flow control is influenced by

- Transmission time: time taken to emit all bits into medium
- Propagation time: time for a bit to traverse the link

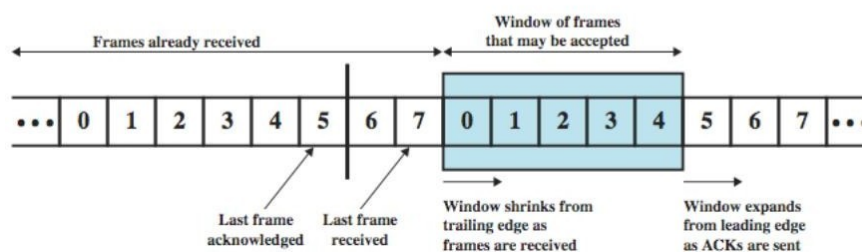
Sliding Windows Flow Control

Sliding Windows Flow Control is a technique used in computer networks to ensure efficient and reliable data transmission between a sender and receiver.

- It allows the transmission of multiple frames.
- If A is the sender and B is the receiver and both will be connected via duplex link.
- Each frame is numbered that is $f_0, f_1, f_2, \dots$
- ACK includes number of next frame expected
Example: RR3 that is received till f_2 and ready to receive f_3 .
- Total number of frames transmitted at a time without waiting for acknowledgement is dependent on size of the field.
- If field size is k , then number starts from 0 to $2^k - 1$.
- If the receiver is not ready to receive the next frame then it can be send the acknowledgement RNR.
Example: RNR5 means we have received till 4th frame but not ready to receive next frame.
- To reopen the window send the normal acknowledgement of RR5.

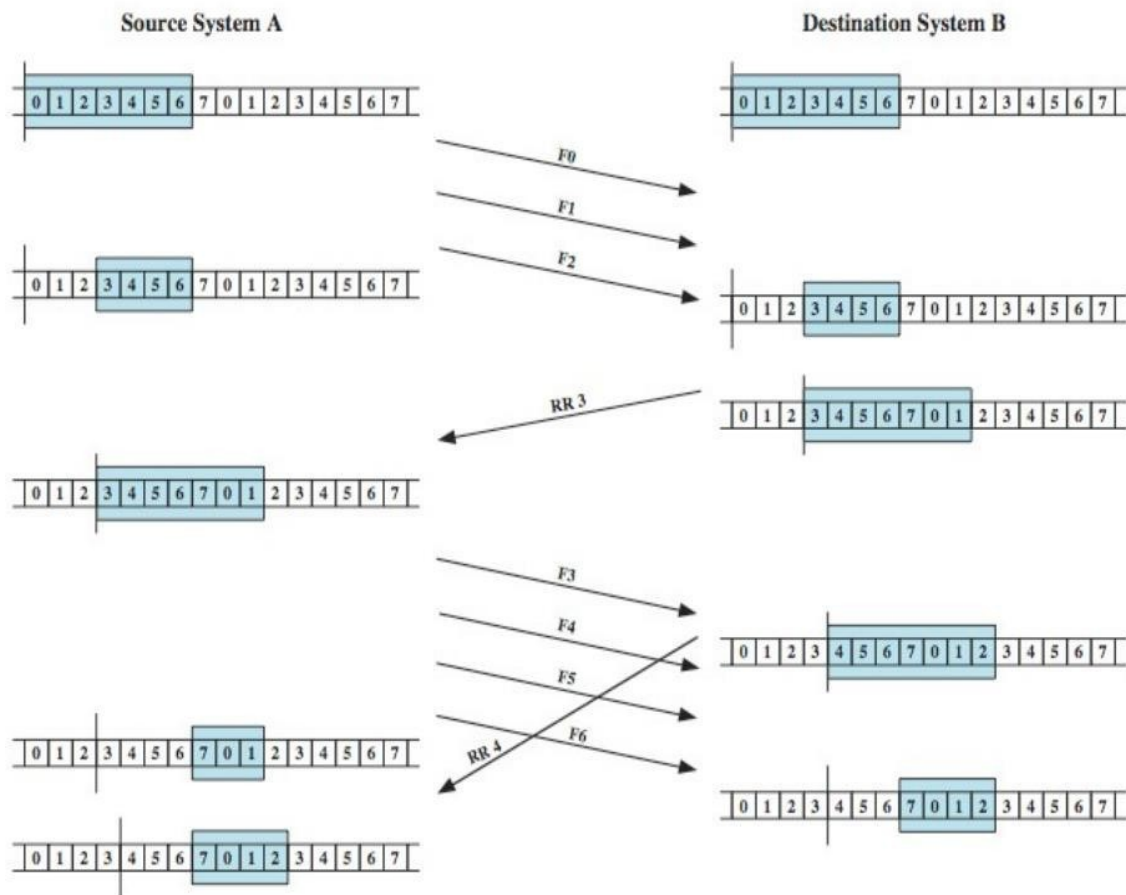


(a) Sender's perspective



(b) Receiver's perspective

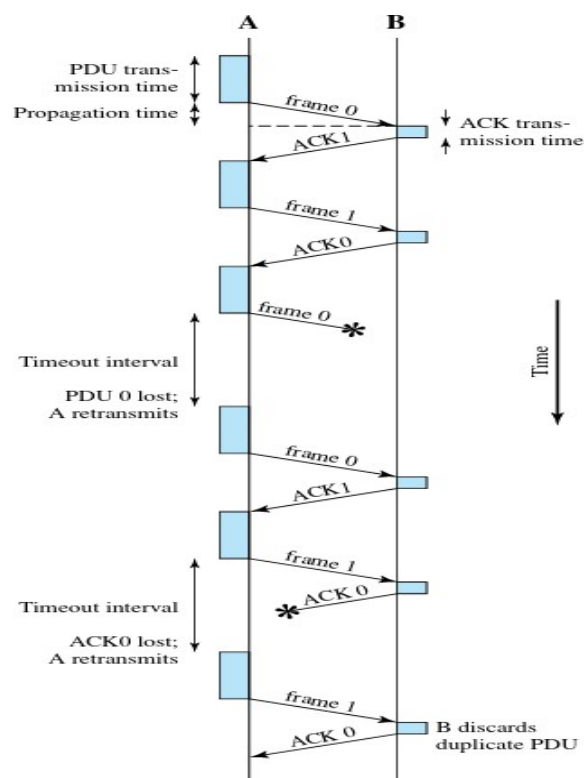
Sliding Window Example:



6. With an example , explain the Stop and Wait ARQ with three bit sequence number for following sequences
- Frame is lost
 - Acknowledgement is lost

Stop-and-wait ARQ is based on the stop-and-wait flow control technique. The source station transmits a single frame and then must await an acknowledgment (ACK). No other data frames can be sent until the destination station's reply arrives at the source station.

- ¬λ If the **frame** is damaged, detect the error and discard it.
 - * Transmitter has timer, if no ACK within timeout, then retransmit frame.
- ¬λ If the **acknowledgement** is damaged, transmitter will not recognize it.
 - * Transmitter will re transmit.
 - * It receives two copies of frame.
 - * To avoid this, use frame0/frame1 and ACK0 / ACK1
 - * Receiver will neglect duplicate copy



The above figure shows the two types of errors just described . The third frame transmitted by A is lost or damaged and therefore B does not return an ACK. A times out and retransmits the frame. Later, A transmits a frame labeled 1 but the ACK0 for that frame is lost. A times out and retransmits the same frame. When B receives two frames in a row with the same label, it discards the second frame but sends back an ACK0 to each.

7. Define Flow Control. Describe stop wait flow control.

Flow control is a technique used to ensure that the sending entity does not overwhelm the receiving entity. This is done by preventing the sending entity from sending more data than the receiving entity can handle.

There are two main factors that influence flow control:

- ¬ Transmission time: This is the time it takes to emit all the bits of a frame into the medium.
- ¬ Propagation time: This is the time it takes for a bit to traverse the link between the sender and receiver.

In the absence of errors, the sender can only send data as fast as the receiver can receive it. This is because the receiver needs to have time to buffer the data before it can be processed.

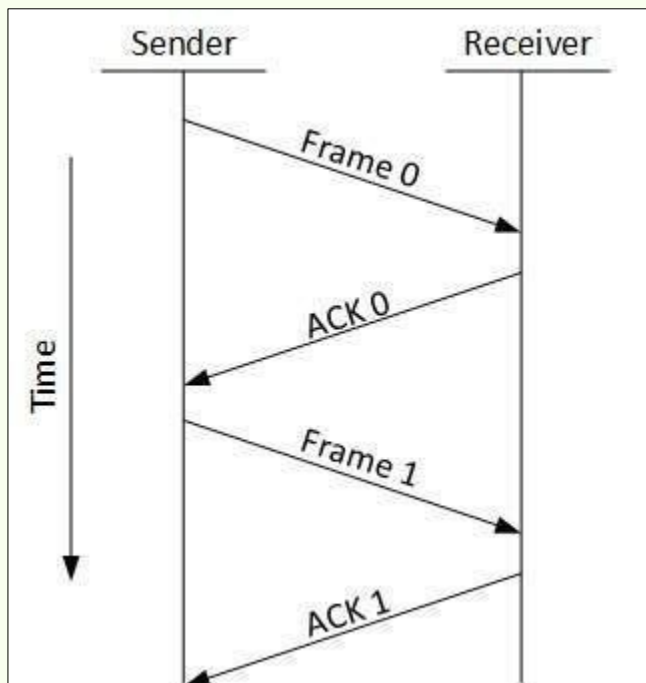
If the sender sends data too quickly, the receiver's buffer will overflow and data will be lost. This is why flow control is important.

Benefits of Flow control:

- ¬ Prevents data loss: Flow control helps to prevent data loss by ensuring that the receiver is able to process all of the data that is sent to it.
- ¬ Improves performance: Flow control can improve performance by allowing the sender to send data at a rate that the receiver can handle. This can help to reduce the amount of time it takes to transmit data.
- ¬ Reduces congestion: Flow control can help to reduce congestion by preventing the sender from sending too much data at once. This can help to ensure that the network is able to handle the data that is being transmitted.

Stop Wait Flow Control

Stop-and-wait flow control is a simple protocol that ensures that data is transmitted reliably and in order. The sender transmits one frame of data at a time and waits for an acknowledgement from the receiver before transmitting the next frame. This ensures that the receiver is able to process all of the data that is sent to it, and that the data is not lost or corrupted.



How stop-and-wait flow control works:

1. The sender sends a frame of data to the receiver.
2. The receiver receives the frame and processes it.
3. The receiver sends an acknowledgement to the sender.
4. The sender sends the next frame of data.
5. Repeat steps 2-4 until all of the data has been transmitted.

If the receiver does not receive a frame correctly, it will send a negative acknowledgement to the sender. The sender will then retransmit the frame.

Stop-and-wait flow control is a simple and reliable protocol, but it is not very efficient. The sender has to wait for an acknowledgement before sending the next frame, which can slow down the transmission of data.

Advantages of stop-and-wait flow control:

- ¬ Reliable: Stop-and-wait flow control ensures that data is transmitted reliably and in order.
- ¬ Simple: Stop-and-wait flow control is a simple protocol to implement.

Disadvantages of stop-and-wait flow control:

- ¬ Inefficient: Stop-and-wait flow control can be inefficient because the sender has to wait for an acknowledgement before sending the next frame.
- ¬ Not scalable: Stop-and-wait flow control is not scalable to high data rates.

8)Write a note on pop3

It allows user transfer agents (on client PCs) to contact the message transfer agent (on the ISP's machine) and allow e-mail to be copied from the ISP to the user.

- POP3 is an abbreviation for Post Office Protocol, whereas '3' refers to the version. Version 3 is the latest and most widely used one.
- It is a simple protocol not having much to offer apart from the download. POP3 downloads an email from the server to one computer (or device) and ends up removing it from the server once it gets downloaded on your computer.
- That means it won't be possible to read your email messages from multiple locations, which is not ideal. The best part of this protocol are that you can read your emails when you are offline.
- And the use of the email server's storage is less. However, as it works with one device and doesn't store the messages on the server, there is a need to backup your computer. Failing to do so will result in you losing all your emails when your computer dies.

There are Four stages of pop3 connection:

1. Authorization (or Authentication):
 - In this stage, the client establishes a connection with the POP3 server.
 - The client sends the login credentials (username and password) to the server for authentication.
 - If the credentials are valid, the server grants access to the client's mailbox.

2. Transaction:

- Once the client is authenticated, it enters the transaction stage where it can issue various commands to interact with the mailbox.
- Common commands include retrieving email headers, downloading complete messages, deleting messages, marking messages as read, and more.
- The server responds to these commands and performs the requested actions.

3. Update:

- After the transaction stage, the server may update the mailbox based on the actions taken by the client.
- For example, if the client deleted messages during the transaction stage, the server marks those messages for deletion. However, they are not immediately removed from the server.

4. Termination:

- In the final stage, the client can issue a termination command to end the POP3 session.
- At this point, the server permanently deletes the messages that were flagged for deletion during the transaction stage.
- The connection between the client and server is closed.

9. What is DNS.Explain?

The DNS results high level human readable names for computers to low level IP addresses.

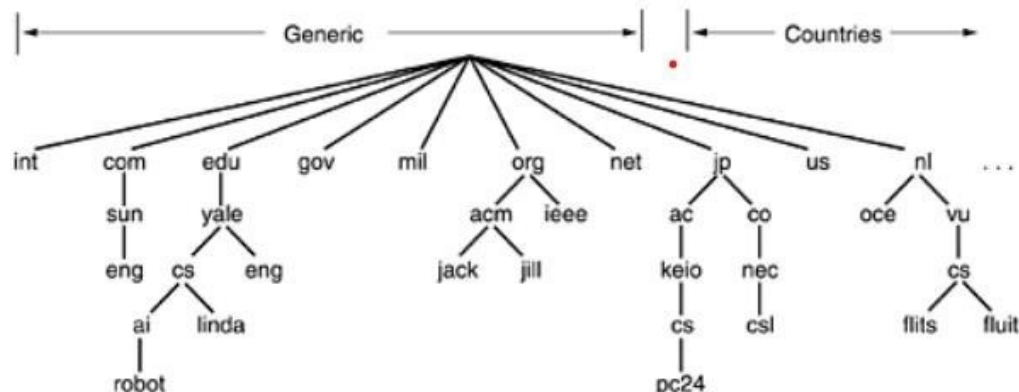
- ↗ **Domain name space**
- ↗ **Domain resource records**
- ↗ **Name servers**

Domain Name

Space:

The Internet is divided into over 200 top-level domains, where each domain covers many hosts. Each domain is partitioned into sub domains, and these are further partitioned, and so on.

All these domains can be represented by a tree. The leaves of the tree represent domains that have no sub domains (but do contain machines, of course). A leaf domain may contain a single host, or it may represent a company and contain thousands of hosts.



The top-level domains come in two flavors: generic and countries. The original generic domains were com (commercial), edu (educational institutions), gov (the U.S. Federal Government), int (certain international organizations), mil (the U.S. armed forces), net (network providers), and org (nonprofit organizations).

In general, getting a second-level domain, such as name-of-company.com, is easy. It merely requires going to a registrar for the corresponding top-level domain (com in this case) to check if the desired name is available and not somebody else's trademark. If there are no problems, the requester pays a small

annual fee and gets the name. By now, virtually every common (English) word has been taken in the com domain.

Each domain is named by the path upward from it to the (unnamed) root. The components are separated by periods (pronounced "dot"). Thus, the engineering department at Sun Microsystems might be **eng.sun.com.**, rather than a UNIX-style name such as **/com/sun/eng.**

Resource Records:

Every domain, whether it is a single host or a top-level domain, can have a set of resource records associated with it. For a single host, the most common resource record is just its IP address, but many other kinds of resource records also exist.

When a resolver gives a domain name to DNS, what it gets back are the resource records associated with that name. Thus, the primary function of DNS is to map domain names onto resource records.

A resource record is a five-tuple. The format we will use is as follows:

Domain_name Time_to_live Class Type Value

The Domain_name tells the domain to which this record applies.

Normally, many records exist for each domain and each copy of the database holds information about multiple domains. This 446 field is thus the primary search key used to satisfy queries. The order of the records in the database is not significant.

The Time_to_live field gives an indication of how stable the record is. Information that is highly stable is assigned a large value, such as 86400 (the number of seconds in 1 day). Information that is highly volatile is assigned a small value, such as 60 (1 minute).

The third field of every resource record is the Class. For Internet information, it is always IN. For non-Internet information, other codes can be used, but in practice, these are rarely seen.

The Type field tells what kind of record this is. The Fig 7-2

Figure 7-2. . The principal DNS resource record types for IPv4.

Type	Meaning	Value
SOA	Start of Authority	Parameters for this zone
A	IP address of a host	32-Bit integer
MX	Mail exchange	Priority, domain willing to accept e-mail
NS	Name Server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
HINFO	Host description	CPU and OS in ASCII
TXT	Text	Uninterpreted ASCII text

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Name Servers

Fighting the IP address for a given host name is called resolution and is done with the DNS Protocol.

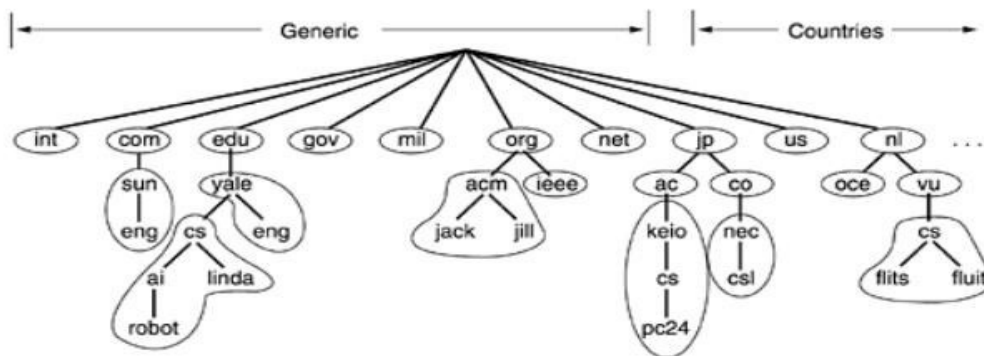
Resolution:

- ↯ Computer requests the local name server to resolve.
- ↯ Local Name Server asks the root name server.
- ↯ Root returns the name server for a lower zone.
- ↯ Continue down zones until name server can answer.

DNS protocol:

- ↯ Runs on UDP 53, retransmits the lost messages.
- ↯ Caches name server answers for better performance.

Figure 7-4. Part of the DNS name space showing the division into zones.

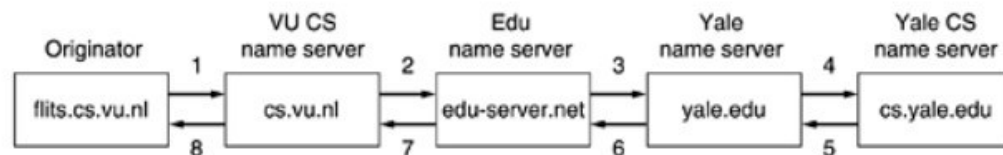


Where the zone boundaries are placed within a zone is up to that zone's administrator. This decision is made in large part based on how many name servers are desired, and where.

For example, in Fig. 7-4, Yale has a server for yale.edu that handles eng.yale.edu but not cs.yale.edu, which is a separate zone with its own name servers. Such a decision might be made when a department

such as English does not wish to run its own name server, but a 449 department such as computer science does. Consequently, cs.yale.edu is a separate zone but eng.yale.edu is not. When a resolver has a query about a domain name, it passes the query to one of the local name servers.

Figure 7-5. How a resolver looks up a remote name in eight steps.



Suppose the local name server has no knowledge of a specific domain and needs to find its address. It may ask nearby name servers, but if they don't know either, it sends a UDP packet to the server for "**edu**" mentioned in its database. That server forwards the request to the name server for "**yale.edu**," which further forwards it to "**cs.yale.edu**." The authoritative resource records are obtained through this process. The records are cached in the **cs.vu.nl** name server but are not authoritative globally. Cache entries should not exist for too long, and the **Time_to_live** field in each resource record determines the caching duration. This method is called a recursive query, where each server looks for the information and reports back. An alternative method is also possible, where the query fails locally, but the name of the next server to try is returned. Some servers don't support recursive queries and always provide the name of the next server to attempt.

While DNS is extremely important to the correct functioning of the Internet, all it really does is map symbolic names for machines onto their IP addresses. It does not help locate people, resources, services, or objects in general. For locating these things, another directory service 450 has been defined, called LDAP (Lightweight Directory Access Protocol). It is a simplified version of the OSI X.500 directory service and is described in RFC 2251

10. Which type Transmission used in Email and Chat. Differentiate between both transmission (Synchronous and Asynchronous)

Email is Asynchronous type of transmission and Chat is Synchronous type of transmission.

Synchronous Transmission:

- ↵ In synchronous transmission, data is sent in a continuous stream or sequence.
- ↵ The sender and receiver must be synchronized and operate at the same clock rate or timing.
- ↵ Data is divided into blocks or frames, and each block is transmitted with the accompanying timing information.
- ↵ Synchronous transmission is typically used for high-speed, reliable data transfer over long distances.
- ↵ Examples of synchronous transmission include synchronous serial communication protocols like Synchronous Optical Networking (SONET) and Synchronous Digital Hierarchy (SDH),Email.
- ↵ synchronous transmission relies on continuous streams of data with synchronized clocks. start and stop bits are not used in transmitting data.
- ↵ Low Latency due to real time communication

Asynchronous Transmission:

- ↵ In asynchronous transmission, data is transmitted in small units known as characters.
- ↵ Each character is typically preceded by a start bit and followed by one or more stop bits to establish synchronization.
- ↵ The timing between characters can vary, and there is no continuous clock signal between the sender and receiver.
- ↵ Asynchronous transmission is commonly used for lower-speed communication and short-distance data transfer.
- ↵ Examples of asynchronous transmission include RS-232 (serial communication) and Universal Asynchronous Receiver-Transmitter (UART) interfaces,Chats.
- ↵ Asynchronous transmission uses individual characters with start and stop bits to establish synchronization.
- ↵ High Latency due to processing time and waiting for data to become available.

11. Explain any 4 types of HTTP Request method ?

HTTP has several request methods.

METHODS	ACTION
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
POST	Sends some information from the client to the server
PUT	Sends a document from the server to the client
TRACE	Echoes the incoming request
CONNECT	Reserved
DELETE	DELETE Remove the Web page
OPTIONS	Enquires about available options

12) Differentiate between POP3 and IMAP

key	POP3	IMAP
Fullform	POP3 stands for Post Office Protocol 3	IMAP stands for Internet Message Access Protocol
Complexity	POP3 is simple and only mails can be downloaded from your inbox to local computer.	IMAP is complex and allows to see all the folders on the mail server.
Ports	POP3 listens on 110 and POP with SSL, POP3DS listens on 995 port.	IMAP listens on 143 and IMAP with SSL, IMAPDS listens on 993 port
Multi access	POP3 supports single device to access the mail at a time.	IMAP supports multiple devices which can access the mail at a time.
Download	In POP3, mail to be downloaded first then can be read.	In IMAP, mail can be partially read before complete download.
Mail Organize	Mails cannot be organized on mail server using POP3.	IMAP allows to organize mails on mail server.
Update Email	Mails cannot be created/updated/deleted on mail server using POP3.	IMAP allows to create/update/delete mails on mail server.
Update Email	Mail content cannot be searched on mail server using POP3. To search, mail to be downloaded first.	Mail content can be searched on mail server using IMAP.
Download	All message are downloaded at once.	Mail message header can be previewed before a message is to be downloaded.

13) What is HDLC.Explain the basic structure of HDLC?

HDLC (High-Level Data Link Control) is a highly significant data link control protocol (ISO 3009, ISO 4335) that serves as the foundation for numerous other protocols.

Basic Characteristics:

To satisfy a variety of applications, HDLC defines three types of stations, two link configurations, and three data transfer modes of operation.

The three station types are

1. Primary station: Responsible for controlling the operation of the link. Frames issued by the primary are called commands.
2. Secondary station: Operates under the control of the primary station. Frames issued by a secondary are called responses. The primary maintains a separate logical link with each secondary station on the line.
3. Combined station: Combines the features of primary and secondary. A combined station may issue both commands and responses.

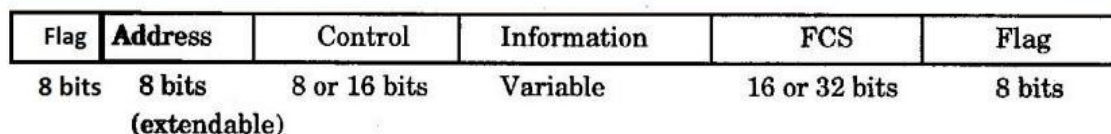
The two link configurations are

1. Unbalanced configuration: Consists of one primary and one or more secondary stations and supports both full-duplex and half-duplex transmission.
2. Balanced configuration: Consists of two combined stations and supports both full-duplex and halfduplex transmission.

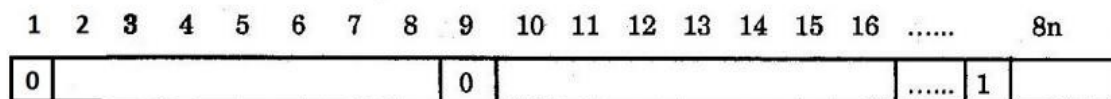
The three data transfer modes are

1. Normal response mode (NRM): Used with an unbalanced configuration. The primary may initiate data transfer to a secondary, but a secondary may only transmit data in response to a command from the primary.
2. Asynchronous balanced mode (ABM): Used with a balanced configuration. Either combined station may initiate transmission without receiving permission from the other combined station.
3. Asynchronous response mode (ARM): Used with an unbalanced configuration. The secondary may initiate transmission without explicit permission of the primary. The primary still retains responsibility for the line, including initialization, error recovery, and logical disconnection.

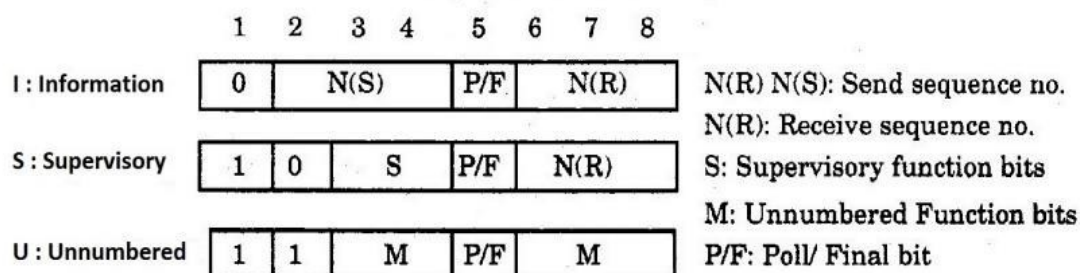
Explain the frame structure of HDLC with block diagram



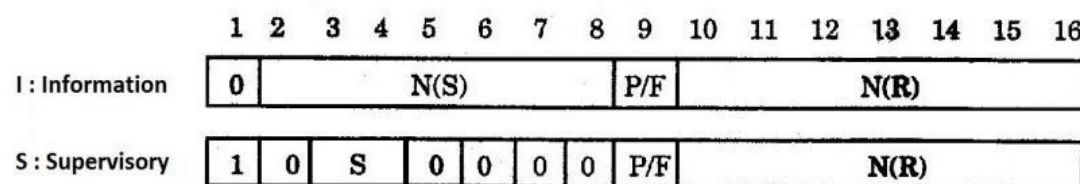
(a) HDLC frame format



(b) Extensible address field



(c) 8-bit control field



(d) 16-bit control field

Flag Field

The flag field is generally responsible for initiation and termination of error checking. In HDLC protocol, there is no start and stop bits. So, the flag field is basically using delimiter 0x7e to simply indicate beginning and end of frame. It is an 8-bit sequence with a bit pattern 01111110 that basically helps in identifying both starting and end of a frame. This bit pattern also serves as a synchronization pattern for receiver. This bit pattern is also not allowed to occur anywhere else inside a complete frame.

Address Field

The address field generally includes HDLC address of secondary station. It helps to identify secondary station will sent or receive data frame. This field also generally consists of 8 bits therefore it is capable of addressing 256 addresses. This field can be of 1 byte or several bytes long, it depends upon requirements of network. Each byte can identify up to 128 stations. This address might include a particular address, a group address, or a broadcast address. A primary address can either be a source of communication or a destination that eliminates requirement of including address of primary.

Control Field

HDLC generally uses this field to determine how to control process of communication. The control field is different for different types of frames in HDLC protocol. The types of frames can be Information frame (I-frame), Supervisory frame (S-frame), and Unnumbered frame (U-frame). This field is a 1-2-byte segment of frame generally requires for flow and error control. This field basically consists of 8 bits but it can be extended to 16 bits. In this field, interpretation of bits usually depends upon the type of frame.

Information Field

This field usually contains data or information of users sender is transmitting to receiver in an I-frame and network layer or management information in U-frame. It also consists of user's data and is fully transparent. The length of this field might vary from one network to another network. Information field is not always present in an HDLC frame.

Frame Check Sequence(FCS)

FCS is generally used for identification of errors i.e., HDLC error detection. In FCS, CRC16 (16-bit Cyclic Redundancy Check) or CRC32 (32-bit Cyclic Redundancy Check) code is basically used for error detection. CRC calculation is done again in receiver. If somehow result differs even slightly from value in original frame, an error is assumed. This field can either contain 2 byte or 4 bytes. This field is a total 16 bit that is required for error detection in address field, control field, and information field. FCS is basically calculated by sender and receiver both of a data frame. FCS is used to confirm and ensure that data frame was not corrupted by medium that is used to transfer frame from sender to receiver.

Three classes of frames are used –

- Information frames (I-Frames) – Carry the actual data. Transport user data from the network layer. In addition, they can also include flow and error control information piggybacked on data.
- Supervisory frames (S-Frames) – Used for error and flow control. They contain, send and receive sequence numbers.
- Unnumbered frames (U-Frames) – Used for various miscellaneous purposes, including link management.